

Measurement Report - Verification of Profile - Demo-Report-Matrix-verify

Created: 2026-08-11 11:21:16

Report Scope

The following profile verification runs are included:

- Demo-Report-Matrix-verify, Instrument: X-Rite i1Pro3 · 1 verification run

No. of Measurements:

1

Date range:

2026-08-10 – 2026-08-10

How to read this report

This report compares what your instrument measured against the chart's design colours (the reference values the chart was built from). Every number is a colour difference (ΔE_{00}): 0 is a perfect match, 1–2 is barely visible, and 10 or more is clearly different.

- Colour accuracy — the ΔE_{00} across the patches, split so you can see the bulk of the chart (all patches and the best 95%) apart from the few hardest patches (the worst 5%). Each metric is judged against your Pass thresholds.
- Paper white & darkest black — the brightest and deepest patches (L^*), a quick health check of your paper and maximum ink.
- Cube corners — paper white, composite black and the six primary and secondary inks. These say as much about your inks as about the instrument.

What the numbers mean depends on how the chart was printed:

- A profiling chart is printed WITHOUT colour management (the raw print you measure to build a profile). It is not expected to match the design closely, so the ΔE can look large — that's normal. Here it is the CHANGE between dated reports that matters, not a single value.
- A verification chart is printed THROUGH your finished profile — ChromIQ converts the sheet itself and prints it with the printer's colour management off. It SHOULD match the design closely, so low ΔE and passes mean the profile is still accurate; rising numbers over time tell you when it's worth re-profiling. (Printed raw instead, the same sheet is a printer drift check — the report says which way each sheet was printed.)

Some of a chart's design colours can be brighter or more saturated than this printer and paper can physically produce — no profile can print them, however good it is. Where the report can tell (it asks the run's profile), it splits the colour-accuracy figures into two groups: "Within the profile's gamut" — the colours that were genuinely printable, the fair measure of accuracy — and "Beyond it" — the unreachable ones, whose distance describes the limit of the gamut, not a mistake of the profile. Every patch stays counted and visible; the Pass/Fail verdict judges the within-gamut figures.

Because the design reference never changes, comparing dated reports of the same chart on the same printer is a clean, reliable signal of drift — ageing inks, a wandering printer, or an instrument going off. Save a report after each measurement to build that history. Screen and print colours here are approximate; the numbers come from your measurement file and are exact.

Report Results

The following results are extracted from the detailed Colour accuracy data shown below for each measurement run. Where a sheet's colours are split into within / beyond the profile's gamut, Pass and Fail judge the within-gamut figures — colours the profile could never print are not counted against it.

Metric	2026-08-10
Average ΔE, all patches	drift
Average ΔE, lowest 95%	drift
Average ΔE, highest 5%	drift
Maximum ΔE, all patches	drift
Maximum ΔE, lowest 95%	drift

Columns marked "drift" are sheets printed raw, without the profile — they are not expected to match the design closely, so Pass and Fail would be unfair to a perfectly healthy printer. For those sheets the detailed chapter shows how far the printer has moved since the previous raw check instead.

Detailed data per measurement run

Measurement run — 2026-08-10T10:00:00 — 105 patches

Profile name: Demo-Report-Matrix-verify

How this verification was produced

What this measured	whether this printer has changed — no profile took part, so this is a drift check, not a profile check
Printed	raw — no profile applied
Colour management at the printer	off — ChromIQ printed the sheet itself
Readings belong to this chart	verified — every patch holds the colour the chart asked for
How the colours were judged	as measured — no white adjustment: every difference counts, the paper's own tone included. (This is a way of comparing, not a rendering intent — a raw print has no intent at all.)
Reference for the ΔE figures	the chart's design colours

Colour accuracy (ΔE_{00} vs the chart's design)

Metric	Within gamut	Beyond it	All patches	Threshold	Result
Average ΔE , all patches	6.03	3.30	5.83	—	—
Average ΔE , lowest 95%	5.64	3.08	5.46	—	—
Average ΔE , highest 5%	13.19	4.82	13.19	—	—
Maximum ΔE , all patches	13.81	4.82	13.81	—	—
Maximum ΔE , lowest 95%	11.01	4.15	11.01	—	—
Spread (std. dev.)	2.72	1.02	2.72	—	—

Drift since the previous raw check (2026-06-01 10:00): average 0.79 ΔE_{00} , maximum 2.85 — this print measured against that print, patch by patch, 105 patches. Small numbers mean your printer still behaves as it did then; growing numbers mean drift — worth re-profiling when they matter to you. (Pass/Fail against the profile thresholds is not shown here: a raw sheet is not expected to match the design closely, so it would fail even a perfectly healthy printer.)

Within what the profile (Demo-Report-Matrix.icc) can print: 97 of this sheet's colours; beyond it: 8. The Result judges the within-gamut figures — a colour beyond the gamut was never printable, so its distance describes the gamut's limit, not a mistake of the profile. Those patches stay visible in their own column, and how steady they are from check to check is a drift signal.

Paper white & darkest black

White (1) — L* 100.0

Black (6) — L* 0.0

Cube corners (the eight ink extremes)

Corner	Expected	Measured	ΔE_{00}
White (1)			13.72
Black (5)			6.45
Red (11)			1.98
Green (13)			3.28
Blue (10)			3.00
Cyan (12)			9.63
Magenta (14)			5.48
Yellow (9)			4.82

Worst patches

Patch	Expected	Measured	$\Delta E00$	Patch	Expected	Measured	$\Delta E00$
2			13.81	62			10.55
4			13.78	24			10.22
1			13.72	96			9.67
3			13.25	12			9.63
100			11.37	45			9.61
47			11.01	33			9.40
85			10.75	59			9.11
73			10.64	105			8.75